BOUNDS ON THE EXCEPTIONAL SET IN THE abc CONJECTURE

C. BERNERT, T. BROWNING, J. D. LICHTMAN, AND J. TERÄVÄINEN

Abstract. We study solutions to the equation $a+b=c$, where a,b,c form a triple of coprime natural numbers. The abc conjecture asserts that, for any $\varepsilon>0$, such triples satisfy $\mathrm{rad}(abc)\geqslant c^{1-\varepsilon}$ with finitely many exceptions. In this article we obtain a power-saving bound on the size of the exceptional set of triples. The proof is based on a combination of upper bounds for the density of integer points on certain high-dimensional varieties, coming from the geometry of numbers and from Fourier analysis.

Contents

1. Introduction

In this paper, we use tools from analytic number theory to estimate the number of triples of a given height satisfying the abc conjecture. Associated to any non-zero integer n is its radical

$$\mathrm{rad}(n)=\prod_{p\mid n}p.$$

We say that a triple $(a,b,c)\in\mathbb{N}^3$ with $\gcd(a,b,c)=1$ is an *abc triple of exponent* λ if

$$a+b=c,\quad \mathrm{rad}(abc)<c^{\lambda}.$$

The well-known abc conjecture of Masser and Oesterlé asserts that, for any $\lambda<1$, there are only finitely many abc triples of exponent λ. See also the work of Robert, Stewart and Tenenbaum [10] for a refinement of the abc conjecture. The best unconditional result is due to Stewart and Yu [9], who have shown that only finitely many abc triples satisfy $\mathrm{rad}(abc)<(\log c)^{3-\varepsilon}$. Recently, Pasten [8] has proved a new subexponential bound, assuming that $a<c^{1-\varepsilon}$, via a connection to Shimura curves. In this paper we shall focus on counting the number $N_\lambda(X)$ of abc triples of exponent λ in a box $[1,X]^3$, as $X\to\infty$.

Given $\lambda>0$, an old result of de Bruijn [3] implies that

$$\#\left\{n\leqslant x:\mathrm{rad}(n)\leqslant x^{\lambda}\right\}=O_\varepsilon(x^{\lambda+\varepsilon}),\tag{1.1}$$

Date: May 12, 2026.

2020 *Mathematics Subject Classification.* 11D45 (11D41, 11D72).

for any $\varepsilon > 0$. Any triple (a, b, c) counted by $N_\lambda(X)$ must satisfy $\mathrm{rad}(abc) < X^\lambda$, and so we must have $\min\{\mathrm{rad}(a)\,\mathrm{rad}(b), \mathrm{rad}(b)\,\mathrm{rad}(c), \mathrm{rad}(c)\,\mathrm{rad}(a)\} < X^{2\lambda/3}$, since a, b, c are pairwise coprime. An application of (1.1) now leads to the following "trivial bound".

Proposition 1.1. *Let $\lambda > 0$. Then $N_\lambda(X) = O_\varepsilon(X^{2\lambda/3+\varepsilon})$, for any $\varepsilon > 0$.*

Note that for $\lambda \geqslant 3$ the counting function $N_\lambda(X)$ is asymptotic to a constant multiple of X^2, as $X \to \infty$, and so it is only worth studying when $\lambda < 3$. The primary goal of this paper is to give the first power-saving improvement over the simple bound in Proposition 1.1 for values of $\lambda \in (0, 2)$. In 2000, it was asked by Mazur [7] whether $N_\lambda(X)$ has exact order $X^{\lambda-1}$, for a fixed $\lambda > 1$. This was answered positively by Kane [6] when $\lambda \geqslant 2$, who used the determinant method to prove that

$$X^{\lambda-1+\varepsilon} \ll N_\lambda(X) \ll_\varepsilon X^{\lambda-1+\varepsilon},$$

for any $\varepsilon > 0$. (In fact, the lower bound is valid for any $\lambda > 1$ and is originally due to unpublished work of Granville.) We assume henceforth that $\lambda \in (0, 2]$. The following is our main result.

Theorem 1.2. *Let $\lambda \in (0, 2]$. For any $\varepsilon > 0$, we have*

$$N_\lambda(X) \ll_\varepsilon X^{\frac{23\lambda+3}{40}+\varepsilon}.$$

In particular, we have $N_\lambda(X) \ll_\lambda X^{0.65}$, if $\lambda \in (0, 1)$.

Our result is non-trivial for a relatively large range of λ, since it follows that

$$N_\lambda(X) = o(X^{2\lambda/3}),$$

for all $\lambda \in (9/11, 2]$. Given the relationship to the *abc* conjecture, the case $\lambda \in (0, 1)$ is of special interest. For such λ, a more refined combination of our bounds leads to a rather complicated optimisation problem, which will yield the following improvement.

Theorem 1.3. *Let $\lambda \in (0, 1)$. Then $N_\lambda(X) \ll_{\varepsilon,\lambda} X^{0.6+\varepsilon}$, for any $\varepsilon > 0$.*

This result is non-trivial for $\lambda \in (9/10, 1)$. The present paper merges an earlier article of the last three authors [2], in which additional bounds coming from the determinant method and the theory of Thue equations were used to produce a worse bound, together with its subsequent improvement and simplification by the first author [1]. Since our bounds do not use the determinant method, it is plausible that further improvements are available, but we expect that a serious numerical improvement will be likely to require new ideas.

Besides our work, we are not aware of any general estimates for $N_\lambda(X)$ when $\lambda < 1$, beyond Proposition 1.1. Nonetheless, there do exist specific Diophantine equations which are covered by the *abc* conjecture and where bounds have been given for the number of solutions. For example, it follows from work of Darmon and Granville [4] that there are only finitely many coprime integer solutions to the Diophantine equation $x^p + y^q = z^r$, when $p, q, r \in \mathbb{N}$ are given and satisfy $1/p + 1/q + 1/r < 1$.

Notation. We shall use $x \sim X$ to denote $x \in (X, 2X]$. All statements involving a number ε are meant to hold for all sufficiently small constants $\varepsilon > 0$. In particular, we will frequently use the "divisor bound" by which we mean the fact that the number of divisors of a positive integer n is $O_\varepsilon(n^\varepsilon)$. For a real number α we write $e(\alpha) = e^{2\pi i\alpha}$ for the corresponding complex number on the unit circle.

Acknowledgements. We are grateful to Bhavik Mehta for his assistance with using Lean to verify parts of an earlier version of this article. While working on this paper the third author was supported by an NSF Postdoctoral Fellowship. The fourth author was supported by the European Union's Horizon Europe research and innovation programme under Marie Skłodowska-Curie grant agreement no. 101058904, by ERC grant agreement no. 101162746, and by Academy of Finland grant no. 362303. This material is based upon work supported by a grant from the Institute for Advanced Study School of Mathematics.

2. An anatomic reduction

Let us denote by $N^*_\lambda(X)$ the number of tuples (a,b,c) counted by $N_\lambda(X)$ satisfying additionally $c \geqslant X/2$ and $\mathrm{rad}(abc) \geqslant c^\lambda/2$. By a dyadic decomposition and the pigeonhole principle, we clearly have

$$N_\lambda(X) \ll (\log X)^2 \max_{1\leqslant Y\leqslant X} \max_{\mu\leqslant\lambda} N^*_\mu(Y).$$

Thus it will suffice to bound $N^*_\lambda(X)$.

The main goal of this section is to bound $N^*_\lambda(X)$ in terms of the number of solutions to certain monomial Diophantine equations. In order to state it, we need to introduce the quantity

$$B_d(a_0,b_0,c_0,\mathbf{A},\mathbf{B},\mathbf{C}) = \#\left\{(\mathbf{a},\mathbf{b},\mathbf{c}) \in \mathbb{N}^{3d} \ : \ \begin{array}{l} a_i \sim A_i,\ b_i \sim B_i,\ c_i \sim C_i, \\ a_0 \prod_{i=1}^d a_i^i + b_0 \prod_{i=1}^d b_i^i = c_0 \prod_{i=1}^d c_i^i, \\ \gcd\left(\prod_{i=0}^d a_i, \prod_{i=0}^d b_i, \prod_{i=0}^d c_i\right) = 1 \end{array}\right\},$$

for $d \in \mathbb{N}$, a vector $(a_0,b_0,c_0) \in \mathbb{N}^3$ and parameters

$$\mathbf{A} = (A_1,\ldots,A_d), \mathbf{B} = (B_1,\ldots,B_d), \mathbf{C} = (C_1,\ldots,C_d) \in \mathbb{N}^d.$$

Bearing this notation in mind, we shall prove the following bound.

Proposition 2.1. *For any $\varepsilon_0 > 0$, there exists a positive integer $d = d(\varepsilon_0)$ such that*

$$N^*_\lambda(X) \ll_{\varepsilon_0} X^{\varepsilon_0} \cdot \max B_d(a_0,b_0,c_0,\mathbf{A},\mathbf{B},\mathbf{C}),$$

where the maximum runs over pairwise coprime $1 \leqslant a_0,b_0,c_0 \leqslant X^{\varepsilon_0}$ and dyadic parameters $A_i,B_i,C_i \in [1,X]$ with

$$\frac{1}{2^{d+1}} X^\lambda \leqslant \prod_{j=1}^d A_j B_j C_j \leqslant X^{\lambda+3\varepsilon_0/4}$$

and

$$\prod_{j=1}^d A_j^j \leqslant X, \quad \prod_{j=1}^d B_j^j \leqslant X, \quad X^{1-\varepsilon_0/2} \leqslant \prod_{j=1}^d C_j^j \leqslant X.$$

This anatomic decomposition of the *abc* equation is based on the following lemma, which provides such a decomposition for a single variable and which can easily be used to recover the bound (1.1).

Lemma 2.2. *For any $\varepsilon_0 > 0$, there exists a positive integer $d = d(\varepsilon_0) \geqslant 6$ such that any positive integer n has a factorization*

$$n = n_0 \prod_{j=1}^d n_j^j$$

with positive integers $n_0, n_1, \dots, n_d$ *such that* $n_1, \dots, n_d$ *are pairwise coprime, and satisfying* $n_0 \leqslant n^{\varepsilon_0}$ *and*

$$n^{-\varepsilon_0} \prod_{j=1}^{d} n_j \leqslant \operatorname{rad}(n) \leqslant n^{\varepsilon_0} \prod_{j=1}^{d} n_j.$$

Proof. We may assume that $\varepsilon_0 < 1$ for otherwise the claim is trivial with $d = 6$. Fix $d = 6\lceil \varepsilon_0^{-2} \rceil$ and $k = \lceil \varepsilon_0^{-1} \rceil$. For a given n, we consider

$$m_j := \prod_{p^j \| n} p,$$

as naive candidates for n_j. We need to modify this construction to be able to truncate our products at $j \leqslant d$. To this end, for $1 \leqslant j \leqslant d$, we define

$$n_j := \begin{cases} m_j & j \neq k \\ m_j \prod_{i>d} m_i^{\lfloor i/k \rfloor} & j = k \end{cases}$$

as well as

$$n_0 := \prod_{j>d} m_j^{j-k\lfloor j/k \rfloor}.$$

Clearly, we have

$$n = \prod_j m_j^j = n_0 \prod_{j=1}^{d} n_j^j$$

and, since the m_i are pairwise coprime, so are $n_1, \dots, n_d$. Moreover,

$$n_0 \leqslant \prod_{j>d} m_j^k \leqslant n^{k/d} \leqslant n^{\varepsilon_0},$$

since $m_j^k \leqslant (m_j^{j/d})^k = (m_j^j)^{k/d}$. Moving on to the radical, we have

$$\operatorname{rad}(n) \leqslant \operatorname{rad}(n_0) \prod_{j=1}^{d} \operatorname{rad}(n_j) \leqslant n^{\varepsilon_0} \prod_{j=1}^{d} n_j.$$

On the other hand, in view of

$$n_k = m_k \prod_{j>d} m_j^{\lfloor j/k \rfloor} \leqslant \left(m_k^k \prod_{j>d} m_j^j \right)^{1/k} \leqslant n^{1/k} \leqslant n^{\varepsilon_0},$$

we find that

$$\operatorname{rad}(n) = \prod_{j \geqslant 1} m_j \geqslant \prod_{j \leqslant d, j \neq k} n_j \geqslant n^{-\varepsilon_0} \prod_{j=1}^{d} n_j,$$

as required. □

Proof of Proposition 2.1. We may assume that X is sufficiently large, for otherwise the claim is trivially true. We will choose $d = d(\varepsilon_0) = 6\lceil (\varepsilon_0/4)^{-2} \rceil$.

Suppose that (a, b, c) is a triple counted by $N_\lambda^*(X)$. By Lemma 2.2 we obtain factorizations

$$a = a_0 \prod_{j=1}^{d} a_j^j, \quad b = b_0 \prod_{j=1}^{d} b_j^j, \quad c = c_0 \prod_{j=1}^{d} c_j^j,$$

with $a_0, b_0, c_0 \leqslant X^{\varepsilon_0/4}$. By assumption, $a + b = c$ and $\gcd(a, b, c) = 1$, so that a, b, c are pairwise coprime and hence also $\prod_{i=0}^{d} a_i$, $\prod_{i=0}^{d} b_i$ and $\prod_{i=0}^{d} c_i$ are pairwise coprime.

Choosing A_i, B_i, C_i to be the unique powers of two such that $a_i \sim A_i, b_i \sim B_i, c_i \sim C_i$, we find that

$$\frac{1}{2^{d+1}} X^\lambda \leqslant \frac{\mathrm{rad}(abc)}{2^d} \leqslant \prod_{j=1}^d A_j B_j C_j \leqslant \prod_{j=1}^d a_j b_j c_j \leqslant \mathrm{rad}(abc) X^{3\varepsilon_0/4} \leqslant X^{\lambda+3\varepsilon_0/4}$$

and

$$\prod_{j=1}^d A_j^j \leqslant \prod_{j=1}^d a_j^j \leqslant X,$$

together with similar upper bounds for $\prod_{j=1}^d B_j^j$ and $\prod_{j=1}^d C_j^j$. For the lower bound, we note that

$$\prod_{j=1}^d C_j^j \geqslant 2^{-d} \prod_{j=1}^d c_j^j \geqslant X^{1-\varepsilon_0/2},$$

provided that X is sufficiently large.

This shows that (a, b, c) is counted by $B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C})$ for a suitable choice of A_i, B_i, C_i as powers of two. Since there are $O(X^{3\varepsilon_0/4})$ choices for a_0, b_0, c_0 and at most $(\log X)^{3d} \ll_{\varepsilon_0} X^{\varepsilon_0/4}$ choices for A_i, B_i, C_i, the claim follows from the pigeonhole principle. □

In the following two sections, we will study bounds for $B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C})$ for given parameters $a_0, b_0, c_0, A_i, B_i, C_i$ satisfying the conditions in Proposition 2.1, for a given value of ε_0 and the corresponding value of $d = d(\varepsilon_0)$, as well as for a sufficiently large parameter X. To simplify notation, let us write

$$A = \prod_{i=1}^d A_i, \quad B = \prod_{i=1}^d B_i, \quad C = \prod_{i=1}^d C_i.$$

Note that by fixing two of the three sets of variables in the defining equation of $B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C})$, the remaining ones are determined up to a divisor function. We may thus record our first estimate, which recovers Proposition 1.1, since $ABC \leqslant X^{\lambda+3\varepsilon_0}$.

Proposition 2.3. *We have*

$$B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}) \ll_\varepsilon X^\varepsilon \min(AB, AC, BC).$$

3. Fourier analysis

In this section, we bound $B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C})$ using Fourier analysis. (Note that our bound also recovers Proposition 1.1.)

Proposition 3.1. *For $2 \leqslant j, k, \ell \leqslant d$, we have*

$$B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}) \ll_{\varepsilon_0} \frac{X^{2\lambda/3+3\varepsilon_0}}{\left(\prod_{j|i} A_i \prod_{k|i} B_i \prod_{\ell|i} C_i\right)^{1/6}}.$$

Proof. By orthogonality of characters and Hölder's inequality, we have

$$B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}) \leqslant \int_0^1 |S_1(\alpha) S_2(\alpha) S_3(-\alpha)| \, \mathrm{d}\alpha \leqslant \left(\prod_{i=1}^3 \int_0^1 |S_i(\alpha)|^3 \, \mathrm{d}\alpha \right)^{1/3},$$

with the exponential sums

$$S_1(\alpha) = \sum_{a_i \sim A_i} e(\alpha a_0 a_1 a_2^2 \cdots a_d^d), \quad S_2(\alpha) = \sum_{b_i \sim B_i} e(\alpha b_0 b_1 b_2^2 \cdots b_d^d)$$

and
$$S_3(\alpha) = \sum_{c_i \sim C_i} e(\alpha c_0 c_1 c_2^2 \cdots c_d^d).$$
We claim that
$$\int_0^1 |S_1(\alpha)|^2 \,\mathrm{d}\alpha \ll_{d,\varepsilon} X^\varepsilon A \tag{3.1}$$
and
$$\int_0^1 |S_1(\alpha)|^4 \,\mathrm{d}\alpha \ll_{d,\varepsilon} X^\varepsilon \frac{A^3}{\prod_{j|i} A_i}. \tag{3.2}$$
Combined with an application of Cauchy-Schwarz and the analogous inequalities for S_2 and S_3, this will suffice to deduce the result.

For the proof of the second moment bound (3.1), we note that, by orthogonality, the second moment counts the number of solutions to $a_1 a_2^2 \cdots a_d^d = a_1' a_2'^2 \cdots a_d'^d$ with $a_i, a_i' \sim A_i$. But on fixing all the a_i, the a_i' are determined up to $O(X^\varepsilon)$ many choices by the divisor bound. This proves (3.1).

Turning to the fourth moment bound (3.2), we begin by applying Cauchy-Schwarz to the sum over a_i for $j \nmid i$ in $S_1(\alpha)$ to obtain
$$|S_1(\alpha)|^2 \leqslant \prod_{j \nmid i} A_i \sum_{a_i \sim A_i} \sum_{a_i' \sim A_i, j|i} e\left(\alpha a_0 \left(\prod_{j|i} a_i^i - \prod_{j|i} (a_i')^i\right) \prod_{j \nmid i} a_i^i\right).$$
Inserting this bound into our fourth moment and using orthogonality of characters, we find that the fourth moment is bounded by $\prod_{j\nmid i} A_i$ times the number of solutions to
$$\left(\prod_{j|i} a_i^i - \prod_{j|i} a_i'^i\right) \prod_{j\nmid i} a_i^i = \prod_i (a_i'')^i - \prod_i (a_i''')^i$$
with $a_i, a_i'', a_i''' \sim A_i$ and, for $j \mid i$, also $a_i' \sim A_i$. It therefore suffices to show that the number of such solutions is bounded by $O(X^\varepsilon A^2)$. This is certainly true for the cases where both sides are equal to zero, since we can choose the a_i and a_i'' freely and then the a_i' and a_i''' are determined up to a divisor function.

In the remaining cases, we have at most A^2 many choices for the a_i'' and a_i'''. But then the left-hand side is fixed. In particular, the a_i for $j \nmid i$ are determined up to a divisor function. We now recall the well-known fact that, for a fixed non-zero integer $n \in \mathbb{Z}$, the Diophantine equation $x^j - y^j = n$ has $O_\varepsilon(|nX|^\varepsilon)$ integer solutions in the box $[1, X]^2$. (This can be proved using the divisor bound.) Thus it follows that a_i and a_i' are determined up to a divisor function. This proves (3.2) and therefore the proposition. □

4. Geometry of numbers

In this section, we bound $B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C})$ by fixing some of the variables and treating the remaining equation as a linear equation via the geometry of numbers. This leads to the following bound.

Proposition 4.1. *For any sets $I, I', I'' \subset \{1, 2, \dots, d\}$ we have*
$$B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}) \ll_{\varepsilon_0} \frac{X^{\lambda + 4\varepsilon_0}}{\prod_{i\in I} A_i \prod_{i \in I'} B_i \prod_{i\in I''} C_i} + X^{\lambda - 1 + 6\varepsilon_0} \prod_{i \in I} A_i^{i-1} \prod_{i\in I'} B_i^{i-1} \prod_{i \in I''} C_i^{i-1}.$$

Proof. Let $(a_1, \dots, a_d, b_1, \dots, b_d, c_1, \dots, c_d)$ be a tuple counted by $B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C})$. Define

$$\begin{aligned} a' &= a_0 \prod_{i \notin I} a_i^i, & b' &= b_0 \prod_{i \notin I'} b_i^i, & c' &= c_0 \prod_{i \notin I''} c_i^i, \\ x &= \prod_{i \in I} a_i^i, & y &= \prod_{i \in I'} b_i^i, & z &= \prod_{i \in I''} c_i^i, \\ X &= \prod_{i \in I} A_i^i, & Y &= \prod_{i \in I'} B_i^i, & Z &= \prod_{i \in I''} C_i^i. \end{aligned}$$

Then $\gcd(a', b', c') = 1$ and $\gcd(x, y, z) = 1$ and we have

$$a'x + b'y = c'z.$$

By a classical bound from the geometry of numbers, as proved by Heath-Brown [5, Lemma 3], for example, for fixed a', b', c', the number of primitive solutions (x, y, z) is bounded by

$$\ll 1 + \frac{XYZ}{\max\{|a'|X, |b'|Y, |c'|Z\}}.$$

Any such solution fixes a_i for $i \in I$, b_i for $i \in I'$ and c_i for $i \in I''$ up to $O(X^\varepsilon)$ many choices. Summing over the choices of the remaining variables, the $O(1)$-term contributes

$$\begin{aligned} \ll_\varepsilon X^\varepsilon \prod_{i \notin I} A_i \prod_{i \notin I'} B_i \prod_{i \notin I''} C_i &= X^\varepsilon \frac{ABC}{\prod_{i \in I} A_i \prod_{i \in I'} B_i \prod_{i \in I''} C_i} \\ &\ll_{\varepsilon_0} \frac{X^{\lambda + 4\varepsilon_0}}{\prod_{i \in I} A_i \prod_{i \in I'} B_i \prod_{i \in I''} C_i}, \end{aligned}$$

since $ABC \leqslant X^{\lambda + 3\varepsilon_0}$, which is satisfactory. Similarly, the remaining term contributes

$$\begin{aligned} &\ll_\varepsilon X^\varepsilon \frac{\prod_{i \in I} A_i^i \prod_{i \in I'} B_i^i \prod_{i \in I''} C_i^i}{X^{1-2\varepsilon}} \cdot \prod_{i \notin I} A_i \prod_{i \notin I'} B_i \prod_{i \notin I''} C_i \\ &\ll_{\varepsilon_0} X^{-1+3\varepsilon_0} ABC \prod_{i \in I} A_i^{i-1} \prod_{i \in I'} B_i^{i-1} \prod_{i \in I''} C_i^{i-1}, \end{aligned}$$

which is also satisfactory. □

5. The simplest non-trivial bound

In this section we prove Theorem 1.2. For convenience let us write $P_i = A_i B_i C_i$.

Proof of Theorem 1.2. Let $\lambda \in (0, 2]$ and suppose that the theorem is false. By Proposition 2.1, we can then find $\delta_0 < \frac{11\lambda - 9}{120}$ such that for any sufficiently small $\varepsilon_0 > 0$, and for the corresponding value of d, there exist $a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}$ with

$$B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}) \geqslant X^{2\lambda/3 - \delta_0},$$

for arbitrarily large X. Fix $\delta \in (\delta_0, \frac{11\lambda - 9}{120})$. For sufficiently small $\varepsilon_0 > 0$, it follows from Proposition 4.1 with $I = I' = I'' = \{1\}$ that we must have $P_1 \ll X^{\lambda/3 + \delta}$. By Proposition 3.1 with $j = k = \ell$, we must have $P_j \ll X^{6\delta}$ for all $j \geqslant 2$. However,

$$X^{5\lambda - 3} \ll \prod_i P_i^{5-i} \ll P_1^4 P_2^3 P_3^2 P_4 \ll X^{4\lambda/3 + 40\delta},$$

which contradicts our upper bound for δ. □

6. A stronger result for $\lambda < 1$

In this section we describe how to prove Theorem 1.3. To begin with, the following result is obtained by solving a mixed-integer linear program; the corresponding `Python` code is given in Section A.

Proposition 6.1. *Let $a, a_1, \dots, a_6, b, b_1, \dots, b_6, c, c_1, \dots, c_6, D \in \mathbb{R}_{\geqslant 0}$. Let*

$$p_i = a_i + b_i + c_i, \quad L = a + b + c.$$

Then $D \leqslant 0.6$ if the following list of 22 inequalities hold:

$$L \leqslant 1, \tag{6.1}$$

$$\begin{cases} a_1 + a_2 + a_3 + a_4 + a_5 + a_6 \leqslant a, \\ b_1 + b_2 + b_3 + b_4 + b_5 + b_6 \leqslant b, \\ c_1 + c_2 + c_3 + c_4 + c_5 + c_6 \leqslant c, \end{cases} \tag{6.2}$$

$$\begin{cases} 7a - 1 \leqslant 6a_1 + 5a_2 + 4a_3 + 3a_4 + 2a_5 + a_6, \\ 7b - 1 \leqslant 6b_1 + 5b_2 + 4b_3 + 3b_4 + 2b_5 + b_6, \\ 7c - 1 \leqslant 6c_1 + 5c_2 + 4c_3 + 3c_4 + 2c_5 + c_6, \end{cases} \tag{6.3}$$

$$D \leqslant \min(a + b, a + c, b + c), \tag{6.4}$$

$$\min(p_1 + p_2 + p_4, 1 - p_2 - 3p_4) \leqslant L - D, \tag{6.5}$$

$$\begin{cases} \min(p_1 + p_2 + a_3, 1 - p_2 - 2a_3) \leqslant L - D, \\ \min(p_1 + p_2 + b_3, 1 - p_2 - 2b_3) \leqslant L - D, \\ \min(p_1 + p_2 + c_3, 1 - p_2 - 2c_3) \leqslant L - D, \end{cases} \tag{6.6}$$

$$\begin{cases} \min(p_1 + p_2 + a_3 + b_3, 1 - p_2 - 2a_3 - 2b_3) \leqslant L - D, \\ \min(p_1 + p_2 + b_3 + c_3, 1 - p_2 - 2b_3 - 2c_3) \leqslant L - D, \\ \min(p_1 + p_2 + c_3 + a_3, 1 - p_2 - 2c_3 - 2a_3) \leqslant L - D, \end{cases} \tag{6.7}$$

$$\begin{cases} \min(p_1 + a_3 + b_2 + c_2, 1 - 2a_3 - b_2 - c_2) \leqslant L - D, \\ \min(p_1 + a_2 + b_3 + c_2, 1 - a_2 - 2b_3 - c_2) \leqslant L - D, \\ \min(p_1 + a_2 + b_2 + c_3, 1 - a_2 - b_2 - 2c_3) \leqslant L - D, \end{cases} \tag{6.8}$$

$$\begin{cases} \min(p_1 + a_2 + b_3 + c_3, 1 - a_2 - 2b_3 - 2c_3) \leqslant L - D, \\ \min(p_1 + a_3 + b_2 + c_3, 1 - 2a_3 - b_2 - 2c_3) \leqslant L - D, \\ \min(p_1 + a_3 + b_3 + c_2, 1 - 2a_3 - 2b_3 - c_2) \leqslant L - D, \end{cases} \tag{6.9}$$

$$\begin{aligned} &\max(a_2 + a_4 + a_6, a_3 + a_6, a_5) + \max(b_2 + b_4 + b_6, b_3 + b_6, b_5) \\ &\qquad + \max(c_2 + c_4 + c_6, c_3 + c_6, c_5) \leqslant 4L - 6D. \end{aligned} \tag{6.10}$$

Proof of Theorem 1.3. We fix a choice of $\lambda \in (0,1)$. Suppose that Theorem 1.3 is false. Then, by Proposition 2.1, we can find $\delta_0 > 0.6$ such that for any ε_0 and the corresponding d, there are $a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}$ with

$$B_d(a_0, b_0, c_0, \mathbf{A}, \mathbf{B}, \mathbf{C}) \geqslant X^{\delta_0}$$

for arbitrarily large X. Fix $D \in (0.6, \delta_0)$.

Write $A = X^a, B = X^b, C = X^c$ as well as $A_i = X^{a_i}, B_i = X^{b_i}, C_i = X^{c_i}$. Then, as we will see below, for sufficiently small $\varepsilon_0 > 0$, the variables $a_i, b_i, c_i, a, b, c, D$ satisfy the hypotheses of Proposition 6.1 contradicting the fact that $D > 0.6$.

Assuming that ε_0 is sufficiently small, it follows that $ABC \leqslant X^{\lambda + 3\varepsilon_0} \leqslant X$. This establishes (6.1). The inequalities in (6.2) follow immediately from the definitions. The inequalities in (6.3) also follow from the definitions and the observation that

$$\sum_{1 \leqslant i \leqslant 6} i a_i \leqslant 1, \quad \sum_{1 \leqslant i \leqslant 6} i b_i \leqslant 1, \quad \sum_{1 \leqslant i \leqslant 6} i c_i \leqslant 1,$$

so that

$$7a - 1 \leqslant 7a - \sum_{1 \leqslant i \leqslant 6} i a_i \leqslant 6a_1 + 5a_2 + 4a_3 + 3a_4 + 2a_5 + a_6.$$

Similarly for the remaining two inequalities in (6.3). The inequality (6.4) is just the trivial bound from Proposition 2.3.

The next set of inequalities originate from the geometry bound in Proposition 4.1 with various choices of I, I', I''. More precisely, (6.5) follows on taking

$$I = I' = I'' = \{1, 2, 4\};$$

the inequalities (6.6) correspond to taking

$$I = I' = \{1, 2\}, I'' = \{1, 2, 3\}$$

and its permutations; the inequalities (6.7) correspond to taking

$$I = \{1, 2\}, I' = I'' = \{1, 2, 3\}$$

and its permutations; the inequalities (6.8) correspond to taking

$$I = I' = \{1, 2\}, I'' = \{1, 3\}$$

and its permutations; and finally the inequalities (6.9) correspond to taking

$$I = \{1, 2\}, I' = I'' = \{1, 3\}$$

and its permutations. The final inequality (6.10) is the Fourier bound from Proposition 3.1. $\square$

Appendix A. Python code for the linear program

```
import pulp

# define problem
prob = pulp.LpProblem("LP_with_OR_constraints", pulp.LpMaximize)

# Variables
p1 = pulp.LpVariable("p1", lowBound=0)
a1 = pulp.LpVariable("a1", lowBound=0)
b1 = pulp.LpVariable("b1", lowBound=0)
c1 = pulp.LpVariable("c1", lowBound=0)
p2 = pulp.LpVariable("p2", lowBound=0)
a2 = pulp.LpVariable("a2", lowBound=0)
b2 = pulp.LpVariable("b2", lowBound=0)
```

```
c2 = pulp.LpVariable("c2", lowBound=0)
p3 = pulp.LpVariable("p3", lowBound=0)
a3 = pulp.LpVariable("a3", lowBound=0)
b3 = pulp.LpVariable("b3", lowBound=0)
c3 = pulp.LpVariable("c3", lowBound=0)
p4 = pulp.LpVariable("p4", lowBound=0)
a4 = pulp.LpVariable("a4", lowBound=0)
b4 = pulp.LpVariable("b4", lowBound=0)
c4 = pulp.LpVariable("c4", lowBound=0)
p5 = pulp.LpVariable("p5", lowBound=0)
a5 = pulp.LpVariable("a5", lowBound=0)
b5 = pulp.LpVariable("b5", lowBound=0)
c5 = pulp.LpVariable("c5", lowBound=0)
p6 = pulp.LpVariable("p6", lowBound=0)
a6 = pulp.LpVariable("a6", lowBound=0)
b6 = pulp.LpVariable("b6", lowBound=0)
c6 = pulp.LpVariable("c6", lowBound=0)
a = pulp.LpVariable("a", lowBound=0)
b = pulp.LpVariable("b", lowBound=0)
c = pulp.LpVariable("c", lowBound=0)
A = pulp.LpVariable("A", lowBound=0)
B = pulp.LpVariable("B", lowBound=0)
C = pulp.LpVariable("C", lowBound=0)
D  = pulp.LpVariable("D")
L = pulp.LpVariable("L", lowBound=0)

# binary variables for OR conditions
z1 = pulp.LpVariable("z1", cat="Binary")
z2 = pulp.LpVariable("z2", cat="Binary")
z3 = pulp.LpVariable("z3", cat="Binary")
z4 = pulp.LpVariable("z4", cat="Binary")
z5 = pulp.LpVariable("z5", cat="Binary")
z6 = pulp.LpVariable("z6", cat="Binary")
z7 = pulp.LpVariable("z7", cat="Binary")
z8 = pulp.LpVariable("z8", cat="Binary")
z9 = pulp.LpVariable("z9", cat="Binary")
z10 = pulp.LpVariable("z10", cat="Binary")
z11 = pulp.LpVariable("z11", cat="Binary")
z12 = pulp.LpVariable("z12", cat="Binary")
z13 = pulp.LpVariable("z13", cat="Binary")
M = 100  # Big-M

# Goal function: Maximal exponent
prob += D

# Setup
prob += p1 == a1+b1+c1
prob += p2 == a2+b2+c2
prob += p3 == a3+b3+c3
prob += p4 == a4+b4+c4
prob += p5 == a5+b5+c5
prob += p6 == a6+b6+c6
prob += L == a + b + c

prob += L <= 1
```

```
prob += a1 + a2 + a3 + a4 + a5 + a6 <= a
prob += b1 + b2 + b3 + b4 + b5 + b6 <= b
prob += c1 + c2 + c3 + c4 + c5 + c6 <= c
prob += 7*a - 1 <= 6*a1 + 5*a2 + 4*a3 + 3*a4 + 2*a5 + a6
prob += 7*b - 1 <= 6*b1 + 5*b2 + 4*b3 + 3*b4 + 2*b5 + b6
prob += 7*c - 1 <= 6*c1 + 5*c2 + 4*c3 + 3*c4 + 2*c5 + c6

# Trivial bounds
prob += D <= b + c
prob += D <= a + c
prob += D <= a + b

# Geometry bounds

# 124, 124, 124
prob += p1 + p2 + p4 <= L - D + M*z1
prob += 1 - p2 - 3*p4 <= L - D + M*(1 - z1)

# 12, 12, 123
prob += p1 + p2 + a3 <= L - D + M*z2
prob += 1 - p2 - 2*a3 <= L - D + M*(1 - z2)
prob += p1 + p2 + b3 <= L - D + M*z3
prob += 1 - p2 - 2*b3 <= L - D + M*(1 - z3)
prob += p1 + p2 + c3 <= L - D + M*z4
prob += 1 - p2 - 2*c3 <= L - D + M*(1 - z4)

# 12, 123, 123
prob += p1 + p2 + a3 + b3 <= L - D + M*z5
prob += 1 - p2 - 2*a3 - 2*b3 <= L - D + M*(1 - z5)
prob += p1 + p2 + b3 + c3 <= L - D + M*z6
prob += 1 - p2 - 2*b3 - 2*c3 <= L - D + M*(1 - z6)
prob += p1 + p2 + c3 + a3 <= L - D + M*z7
prob += 1 - p2 - 2*c3 - 2*a3 <= L - D + M*(1 - z7)

# 12,12,13
prob += p1 + a2 + b3 + c2 <= L - D + M*z8
prob += 1 - a2 - 2*b3 - c2 <= L - D + M*(1 - z8)
prob += p1 + a2 + c3 + b2 <= L - D + M*z9
prob += 1 - a2 - 2*c3 - b2 <= L - D + M*(1 - z9)
prob += p1 + b2 + a3 + c2 <= L - D + M*z10
prob += 1 - b2 - 2*a3 - c2 <= L - D + M*(1 - z10)

# 12,13,13
prob += p1 + b2 + c3 + a3 <= L - D + M*z11
prob += 1 - b2 - 2*c3 - 2*a3 <= L - D + M*(1 - z11)
prob += p1 + c2 + a3 + b3 <= L - D + M*z12
prob += 1 - c2 - 2*a3 - 2*b3 <= L - D + M*(1 - z12)
prob += p1 + a2 + b3 + a3 <= L - D + M*z13
prob += 1 - a2 - 2*b3 - 2*c3 <= L - D + M*(1 - z13)

# auxiliary variables for Fourier bound: A=max(a2+a4+a6,a3+a6,a5)
prob += a2 + a4 + a6 <= A
prob += a3 + a6 <= A
prob += a5 <= A
prob += b2 + b4 + b6 <= B
```

```
prob += b3 + b6 <= B
prob += b5 <= B
prob += c2 + c4 + c6 <= C
prob += c3 + c6 <= C
prob += c5 <= C

# Fourier bound
prob += A + B + C <= 4*L - 6*D

# Solve
prob.solve(pulp.PULP_CBC_CMD(msg=False))

# Output
print("Status:", pulp.LpStatus[prob.status])
for v in [a1,b1,c1,a2,b2,c2,a3,b3,c3,a4,b4,c4,a5,b5,c5,a6,b6,c6,D]:
    print(f"{v.name} = {v.value()}")

print("Maximal value of D:", pulp.value(prob.objective))
```

References

[1] C. Bernert, The exceptional set in the abc conjecture. *Preprint*, 2025. (arXiv:2506.13364v1) 2

[2] T. Browning, J. D. Lichtman and J. Teräväinen, The exceptional set in the abc conjecture. *Preprint*, 2024. (arXiv:2410.12234) 2

[3] N.G. de Bruijn, On the number of integers $\leqslant x$ whose prime factors divide n. *Illinois J. Math.* **6** (1962), 137–141. 1

[4] H. Darmon and A. Granville, On the equations $z^m = F(x,y)$ and $Ax^p + By^q = Cz^r$. *Bull. London Math. Soc.* **27** (1995), 513–543. 2

[5] D.R. Heath-Brown, Diophantine approximation with square-free numbers. *Math. Zeit.* **187** (1984), 335–344. 7

[6] D. Kane, On the number of ABC solutions with restricted radical sizes. *J. Number Theory* **154** (2015), 32–43. 2

[7] B. Mazur, Questions about powers of numbers. *Notices Amer. Math. Soc.* **47** (2000), 195–202. 2

[8] H. Pasten, The largest prime factor of n^2+1 and improvements on subexponential ABC. *Invent. Math.* **236** (2024), 373–385. 1

[9] C.L. Stewart and K.R. Yu, On the *abc* conjecture. II. *Duke Math. J.* **108** (2001), 169–181. 1

[10] O. Robert, C.L. Stewart and G. Tenenbaum, A refinement of the *abc* conjecture. *Bull. London Math. Soc.* **46** (2014), 1156–1166. 1

Institute for Science and Technology Austria, Am Campus 1, 3400 Klosterneuburg, Austria

Email address: christian.bernert@ist.ac.at, tdb@ist.ac.at

Department of Mathematics, Stanford University, 450 Jane Stanford Way, Stanford, CA 94305-2125, USA

Email address: jdlichtman@stanford.edu

Department of Pure Mathematics and Mathematical Statistics, University of Cambridge, Cambridge CB3 0WB, UK

Email address: joni.p.teravainen@gmail.com